

Global CO₂ Emissions: Trends, Regional Shifts, and Projections

Historical analysis spanning 1750–2024

A data-driven overview of how global carbon dioxide emissions have evolved from the Industrial Revolution to 2024, highlighting regional variations and future projections.

Source: Global Carbon Budget 2025, NASA Climate Observatory

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

Current level (Feb 2026)

~280 ppm

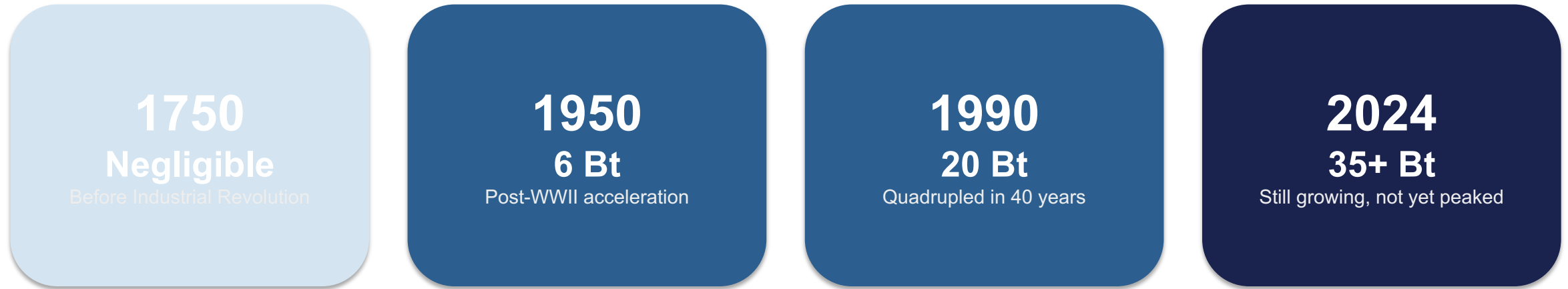
Pre-industrial baseline

Key Context

- 50%+ increase from pre-industrial levels
- 1.5× the natural rise at end of last ice age
- Change took millennia then, centuries now
- Driven by coal, oil, and natural gas burning
- Confirmed by ground and satellite measurements

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

Reference: figure_2.jpg (emissions trajectory)



While emissions growth has slowed in recent years, they have not yet peaked. The trajectory from 6 Bt in 1950 to over 35 Bt today represents a six-fold increase in roughly seven decades.

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

Reference: figure_3.jpg (annual emissions by country)

Country/Region	Annual CO ₂ (Bt)	Share of Global	Trend
China	~12.5	~27%	Slight increase
United States	~5.0	~14%	Declining
India	~3.0	~8%	Rising
EU-27	~2.7	~7%	Declining
Germany	~0.6	~1.7%	Declining
United Kingdom	~0.3	~0.9%	Declining
Brazil	~0.5	~1.4%	Stable
France	~0.3	~0.8%	Declining

China produces over one-quarter of global CO₂ — more than the US, EU-27, India, and Japan combined.

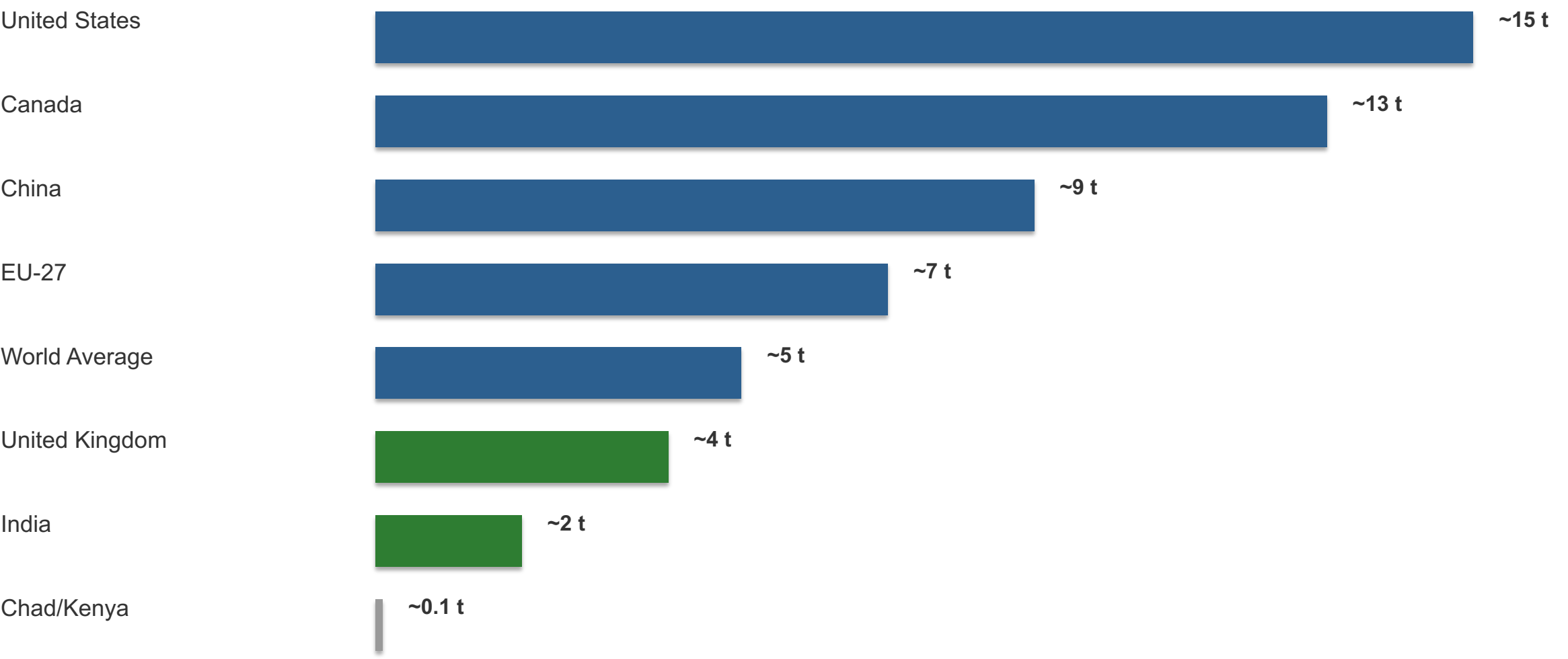
Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third



The geographic center of emissions has shifted dramatically — from >90% in Europe and the US in 1900 to Asia producing

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

Reference: figure_1.jpg (per capita emissions over time)



The average American emits in under 2 days what the average person in Chad or Niger emits in an entire year.

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Reference: figure_1.jpg (annual percentage change map)

DECLINING (Blue)

-1% to -5%

- United States
- Much of Europe
- Some South American nations

INCREASING (Orange/Red)

+2% to +10%+

- Russia
- Southeast Asia
- Several African nations
- Some South American countries (>10%)

The 2024 map reveals a complex picture: while the US and Europe are reducing emissions, growth in Russia, Southeast Asia, and parts of Africa and South America is offsetting these gains.

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita emissions

Country/Region	Primary Energy Sources	Per Capita CO ₂ (t)
France	High nuclear + renewables (~75% nuclear)	~4
United Kingdom	Natural gas, renewables, some coal	~4
Germany	Higher fossil fuel share (~50% coal)	~8
Netherlands	Natural gas, some coal, growing renewables	~7

Policy and energy mix choices — not just prosperity — determine per capita outcomes. France and the UK have much lower per capita emissions than Germany and the Netherlands despite similar development levels.

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

1

Global CO₂ emissions now exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth rates.

2

China alone accounts for over 25% of global emissions (~12.5 Bt), more than double the United States (~5 Bt); India (~3 Bt) is the third-largest and rising fast.

3

The geographic center of emissions has shifted dramatically: from >90% in Europe and the US in 1900 to Asia producing ~50% today, while the US and Europe combined are below one-third.

4

Per capita inequalities are extreme — a 150× gap separates the highest emitters (US, Australia at ~15 t/person) from the lowest (Chad, Niger at ~0.1 t/person).

5

Energy mix and policy choices matter: countries with similar prosperity levels (e.g., France vs. Germany) can have very different per capita emissions depending on their reliance on nuclear, renewables, or fossil fuels.